

CSC435 Extra Credit Assignment - I

Chameleon resource used: compute_cascadelake_r

Containers

First I'll answer the questions and then I'll write commands/paste the command(s) images.

1. Container technology used: LXD

- a. Command for LXD installation: `sudo snap install lxd`

2. 5 containers creation:

- a. `sudo lxc launch ubuntu:22.04 server --config limits.cpu=4 --config limits.memory=2GB -- network nat_tasting`
- b. `for i in {1..4}; do
sudo lxc launch ubuntu:22.04 client$i --config limits.cpu=2 --config limits.memory=2GB -- network nat_tasting
done`

3. All containers on Same network:

- a. As you can see from the command 2a and 2b in the end I'm adding nat_testing network, as it is same for all containers, every container is connected to the same network.

4. Copying the datasets to the clients:

- a. One example of copying is below.
- i. winscp@instance: `scp -r /mnt/d/Depaul_Current/Distributed_Systems_1/Assignment_2/_Work /csc435-pa2-KedarChandulur/datasets/Zipped/Dataset2_First_Half.zip cc@129.114.109.50:~/my_mounting_point/temp/`
- ii. chameleon@instance: `lxc file push ~/my_mounting_point/temp/Dataset2_First_Half.zip client1/OG_Dataset/`
- iii. client1@instance: `unzip Dataset2_First_Half.zip Dataset2`

5. OPTIONAL: Running the programs on the containers: Not done.

6. Most challenging aspect of configuring containers: Most challenging aspect was connecting to the containers to the network and also figuring out on how to move the datasets.

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From here, I'm posting all the screen shots of the commands

```
cc@pa4-kedar-extracredit:~$ sudo snap refresh lxd --channel=latest/stable
lxd 5.20-f3dd836 from Canonical✓ refreshed
cc@pa4-kedar-extracredit:~$ snap list
Name      Version      Rev      Tracking      Publisher      Notes
core20    20231123     2105     latest/stable canonical✓     base
core22    20240111     1122     latest/stable canonical✓     base
lxd       5.20-f3dd836 27049    latest/stable canonical✓     -
snapd     2.61.1       20671    latest/stable canonical✓     snapd
cc@pa4-kedar-extracredit:~$ sudo snap refresh lxd --channel=latest/stable
snap "lxd" has no updates available
```

```
cc@pa4-kedar-extracredit:~$ lxd init
Would you like to use LXD clustering? (yes/no) [default=no]:
Do you want to configure a new storage pool? (yes/no) [default=yes]:
Name of the new storage pool [default=default]: lxd_storage_pool
Name of the storage backend to use (zfs, btrfs, ceph, dir, lvm) [default=zfs]:
Create a new ZFS pool? (yes/no) [default=yes]:
Would you like to use an existing empty block device (e.g. a disk or partition)? (yes/no) [default=no]:
Size in GiB of the new loop device (1GiB minimum) [default=30GiB]:
Would you like to connect to a MAAS server? (yes/no) [default=no]:
Would you like to create a new local network bridge? (yes/no) [default=yes]:
What should the new bridge be called? [default=lxdbr0]: lxd_localbridge
What IPv4 address should be used? (CIDR subnet notation, "auto" or "none") [default=auto]: auto
What IPv6 address should be used? (CIDR subnet notation, "auto" or "none") [default=auto]: none
Would you like the LXD server to be available over the network? (yes/no) [default=no]: yes
Address to bind LXD to (not including port) [default=all]: all
Port to bind LXD to [default=8443]: 8443
Would you like stale cached images to be updated automatically? (yes/no) [default=yes]: no
Would you like a YAML "lxd init" preseed to be printed? (yes/no) [default=no]: yes
```

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```
config:
  core.https_address: '[:,]:8443'
  images.auto_update_interval: "0"
networks:
- config:
    ipv4.address: auto
    ipv6.address: none
    description: ""
    name: lxd_localbridge
    type: ""
    project: default
storage_pools:
- config:
    size: 30GiB
    description: ""
    name: lxd_storage_pool
    driver: zfs
profiles:
- config: {}
  description: ""
  devices:
    eth0:
      name: eth0
      network: lxd_localbridge
      type: nic
    root:
      path: /
      pool: lxd_storage_pool
      type: disk
  name: default
projects: []
cluster: null
```

```
cc@pa4-kedar-extracredit:~$ lxc list
```

NAME	STATE	IPV4	IPV6	TYPE	SNAPSHOTS
------	-------	------	------	------	-----------

```
cc@pa4-kedar-extracredit:~$ lxc network list
```

NAME	TYPE	MANAGED	IPV4	IPV6	DESCRIPTION	USED BY	STATE
eno1np0	physical	NO				0	
eno2np1	physical	NO				0	
lxd_localbridge	bridge	YES	10.3.50.1/24	none		1	CREATED
nat_testing	bridge	YES	10.12.78.1/24	fd42:e918:d927:8b1::1/64		0	CREATED

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```
cc@pa4-kedar-extracredit:~$ lxc network info lxd_localbridge
Name: lxd_localbridge
MAC address: 00:16:3e:e2:a6:fb
MTU: 1500
State: up
Type: broadcast
```

```
IP addresses:
  inet 10.3.50.1/24 (global)
```

```
Network usage:
  Bytes received: 0B
  Bytes sent: 0B
  Packets received: 0
  Packets sent: 0
```

```
Bridge:
  ID: 8000.00163ee2a6fb
  STP: false
  Forward delay: 1500
  Default VLAN ID: 1
  VLAN filtering: true
  Upper devices:
```

```
cc@pa4-kedar-extracredit:~$ lxc network info nat_testing
Name: nat_testing
MAC address: 00:16:3e:71:b1:8c
MTU: 1500
State: up
Type: broadcast
```

```
IP addresses:
  inet 10.12.78.1/24 (global)
  inet6 fd42:e918:d927:8b1::1/64 (global)
```

```
Network usage:
  Bytes received: 0B
  Bytes sent: 0B
  Packets received: 0
  Packets sent: 0
```

```
Bridge:
  ID: 8000.00163e71b18c
  STP: false
  Forward delay: 1500
  Default VLAN ID: 1
  VLAN filtering: true
  Upper devices:
```

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```
cc@pa4-kedar-extracredit:~$ ip address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eno1np0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether bc:97:e1:c4:c3:f0 brd ff:ff:ff:ff:ff:ff
    altname enp25s0f0np0
    inet 10.52.0.185/22 metric 100 brd 10.52.3.255 scope global dynamic eno1np0
        valid_lft 82262sec preferred_lft 82262sec
    inet6 fe80::be97:e1ff:fec4:c3f0/64 scope link
        valid_lft forever preferred_lft forever
3: eno2np1: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether bc:97:e1:c4:c3:f1 brd ff:ff:ff:ff:ff:ff
    altname enp25s0f1np1
4: nat_testing: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default qlen 1000
    link/ether 00:16:3e:71:b1:8c brd ff:ff:ff:ff:ff:ff
    inet 10.12.78.1/24 scope global nat_testing
        valid_lft forever preferred_lft forever
    inet6 fd42:e918:d927:8b1::1/64 scope global
        valid_lft forever preferred_lft forever
5: lxd_localbridge: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default qlen 1000
    link/ether 00:16:3e:e2:a6:fb brd ff:ff:ff:ff:ff:ff
    inet 10.3.50.1/24 scope global lxd_localbridge
        valid_lft forever preferred_lft forever
```

```
cc@pa4-kedar-extracredit:~$ lxc network set lxd_localbridge ipv6.firewall false
lxc network set nat_testing ipv4.firewall false
```

```
cc@pa4-kedar-extracredit:~$ sudo firewall-cmd --zone=trusted --change-interface=lxd_localbridge --permanent
sudo firewall-cmd --zone=trusted --change-interface=nat_testing --permanent
sudo firewall-cmd --reload
success
success
success
```

```
cc@pa4-kedar-extracredit:~$ sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
```

```
cc@pa4-kedar-extracredit:~$ sudo ufw allow in on lxd_localbridge
Skipping adding existing rule
Rule added (v6)
cc@pa4-kedar-extracredit:~$ sudo ufw route allow in on lxd_localbridge
sudo ufw route allow out on lxd_localbridge

sudo ufw allow in on nat_testing
sudo ufw route allow in on nat_testing
sudo ufw route allow out on nat_testing
Skipping adding existing rule
Rule added (v6)
Skipping adding existing rule
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Rule added (v6)
```

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```
cc@pa4-kedar-extracredit:~$ sudo ufw status
Status: active
```

To	Action	From
22	ALLOW	Anywhere
Anywhere on lxd_localbridge	ALLOW	Anywhere
Anywhere on nat_testing	ALLOW	Anywhere
22 (v6)	ALLOW	Anywhere (v6)
Anywhere (v6) on lxd_localbridge	ALLOW	Anywhere (v6)
Anywhere (v6) on nat_testing	ALLOW	Anywhere (v6)
80	ALLOW OUT	Anywhere
80 (v6)	ALLOW OUT	Anywhere (v6)
Anywhere	ALLOW FWD	Anywhere on lxd_localbridge
Anywhere on lxd_localbridge	ALLOW FWD	Anywhere
Anywhere	ALLOW FWD	Anywhere on nat_testing
Anywhere on nat_testing	ALLOW FWD	Anywhere
Anywhere (v6)	ALLOW FWD	Anywhere (v6) on lxd_localbridge
Anywhere (v6) on lxd_localbridge	ALLOW FWD	Anywhere (v6)
Anywhere (v6)	ALLOW FWD	Anywhere (v6) on nat_testing
Anywhere (v6) on nat_testing	ALLOW FWD	Anywhere (v6)

```
cc@pa4-kedar-extracredit:~$ sudo lxc launch ubuntu:22.04 server --config limits.cpu=4 --config limits.memory=2GB --network nat_testing
sudo lxc launch ubuntu:22.04 client1 --config limits.cpu=2 --config limits.memory=2GB --network nat_testing
sudo lxc launch ubuntu:22.04 client2 --config limits.cpu=2 --config limits.memory=2GB --network nat_testing
sudo lxc launch ubuntu:22.04 client3 --config limits.cpu=2 --config limits.memory=2GB --network nat_testing
sudo lxc launch ubuntu:22.04 client4 --config limits.cpu=2 --config limits.memory=2GB --network nat_testing
Creating server
Starting server
Creating client1
Starting client1
Creating client2
Starting client2
Creating client3
Starting client3
Creating client4
Starting client4
```

```
cc@pa4-kedar-extracredit:~$ sudo lxc list
```

NAME	STATE	IPV4	IPV6	TYPE	SNAPSHOTS
client1	RUNNING	10.12.78.218 (eth0)	fd42:e918:d927:8b1:216:3eff:fea7:a951 (eth0)	CONTAINER	0
client2	RUNNING	10.12.78.239 (eth0)	fd42:e918:d927:8b1:216:3eff:feeb:23f8 (eth0)	CONTAINER	0
client3	RUNNING	10.12.78.194 (eth0)	fd42:e918:d927:8b1:216:3eff:fe3f:324 (eth0)	CONTAINER	0
client4	RUNNING	10.12.78.207 (eth0)	fd42:e918:d927:8b1:216:3eff:feea:417b (eth0)	CONTAINER	0
server	RUNNING	10.12.78.143 (eth0)	fd42:e918:d927:8b1:216:3eff:fe0b:6699 (eth0)	CONTAINER	0

```
cc@pa4-kedar-extracredit:~$ sudo lxc network list
```

NAME	TYPE	MANAGED	IPV4	IPV6	DESCRIPTION	USED BY	STATE
eno1np0	physical	NO				0	
eno2np1	physical	NO				0	
lxd_localbridge	bridge	YES	10.3.50.1/24	none		1	CREATED
nat_testing	bridge	YES	10.12.78.1/24	fd42:e918:d927:8b1::1/64		5	CREATED

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```
cc@pa4-kedar-extracredit:~$ lxc exec server -- free -m
              total        used        free      shared  buff/cache   available
Mem:           1907          82        1707          0         116        1824
Swap:            0            0            0
cc@pa4-kedar-extracredit:~$ lxc exec server -- nproc
4
```

```
cc@pa4-kedar-extracredit:~$ lxc exec client1 -- free -m
              total        used        free      shared  buff/cache   available
Mem:           1907          82        1708          0         116        1824
Swap:            0            0            0
cc@pa4-kedar-extracredit:~$ lxc exec client1 -- nproc
2
```

```
cc@pa4-kedar-extracredit:~$ lxc exec server ping google.com
PING google.com (142.250.113.102) 56(84) bytes of data.
64 bytes from rs-in-f102.1e100.net (142.250.113.102): icmp_seq=1 ttl=106 time=7.11 ms
64 bytes from rs-in-f102.1e100.net (142.250.113.102): icmp_seq=2 ttl=106 time=7.10 ms
^C
--- google.com ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 7.098/7.103/7.108/0.005 ms
cc@pa4-kedar-extracredit:~$ lxc exec client1 ping google.com
PING google.com (142.250.113.138) 56(84) bytes of data.
64 bytes from rs-in-f138.1e100.net (142.250.113.138): icmp_seq=1 ttl=58 time=7.31 ms
64 bytes from rs-in-f138.1e100.net (142.250.113.138): icmp_seq=2 ttl=58 time=7.26 ms
64 bytes from rs-in-f138.1e100.net (142.250.113.138): icmp_seq=3 ttl=58 time=7.27 ms
^C
--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 7.260/7.281/7.314/0.023 ms
```

```
cc@pa4-kedar-extracredit:~$ lxc exec client1 bash
root@client1:~# ip addr
```

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```
root@client1:~# ping 10.12.78.239
PING 10.12.78.239 (10.12.78.239) 56(84) bytes of data.
64 bytes from 10.12.78.239: icmp_seq=1 ttl=64 time=0.113 ms
64 bytes from 10.12.78.239: icmp_seq=2 ttl=64 time=0.040 ms
64 bytes from 10.12.78.239: icmp_seq=3 ttl=64 time=0.040 ms
64 bytes from 10.12.78.239: icmp_seq=4 ttl=64 time=0.040 ms
^C
--- 10.12.78.239 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3076ms
rtt min/avg/max/mdev = 0.040/0.058/0.113/0.031 ms
root@client1:~# ping client2
PING client2(client2.lxd (fd42:e918:d927:8b1:216:3eff:feeb:23f8)) 56 data bytes
64 bytes from client2.lxd (fd42:e918:d927:8b1:216:3eff:feeb:23f8): icmp_seq=1 ttl=64 time=0.131 ms
64 bytes from client2.lxd (fd42:e918:d927:8b1:216:3eff:feeb:23f8): icmp_seq=2 ttl=64 time=0.047 ms
64 bytes from client2.lxd (fd42:e918:d927:8b1:216:3eff:feeb:23f8): icmp_seq=3 ttl=64 time=0.047 ms
64 bytes from client2.lxd (fd42:e918:d927:8b1:216:3eff:feeb:23f8): icmp_seq=4 ttl=64 time=0.047 ms
64 bytes from client2.lxd (fd42:e918:d927:8b1:216:3eff:feeb:23f8): icmp_seq=5 ttl=64 time=0.048 ms
^C
--- client2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 0.047/0.064/0.131/0.033 ms
root@client1:~# ping server
PING server(server.lxd (fd42:e918:d927:8b1:216:3eff:fe0b:6699)) 56 data bytes
64 bytes from server.lxd (fd42:e918:d927:8b1:216:3eff:fe0b:6699): icmp_seq=1 ttl=64 time=0.132 ms
64 bytes from server.lxd (fd42:e918:d927:8b1:216:3eff:fe0b:6699): icmp_seq=2 ttl=64 time=0.047 ms
64 bytes from server.lxd (fd42:e918:d927:8b1:216:3eff:fe0b:6699): icmp_seq=3 ttl=64 time=0.047 ms
^C
--- server ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 0.047/0.075/0.132/0.040 ms
```

```
root@client1:~# sudo apt install ssh
```

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```
root@client1:~# sudo apt install ufw
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ufw is already the newest version (0.36.1-4ubuntu0.1).
ufw set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 3 not upgraded.
root@client1:~# sudo apt-get install openssh-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openssh-server is already the newest version (1:8.9p1-3ubuntu0.6).
0 upgraded, 0 newly installed, 0 to remove and 3 not upgraded.
root@client1:~# sudo service status ssh
status: unrecognized service
root@client1:~# sudo service ssh status
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
   Active: active (running) since Fri 2024-03-08 00:56:38 UTC; 31min ago
     Docs: man:sshd(8)
           man:sshd_config(5)
    Main PID: 355 (sshd)
      Tasks: 1 (limit: 230129)
     Memory: 2.6M
        CPU: 18ms
    CGroup: /system.slice/ssh.service
            └─355 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Mar 08 00:56:38 client1 systemd[1]: Starting OpenBSD Secure Shell server...
Mar 08 00:56:38 client1 sshd[355]: Server listening on 0.0.0.0 port 22.
Mar 08 00:56:38 client1 sshd[355]: Server listening on :: port 22.
Mar 08 00:56:38 client1 systemd[1]: Started OpenBSD Secure Shell server.
root@client1:~# sudo ufw allow 22/tcp
Rules updated
Rules updated (v6)
```

```
root@client1:/OG_Dataset# ls
Dataset1 Dataset2 Dataset3 Dataset4 Dataset5
root@client1:/OG_Dataset# ls Dataset1
folder1 folder11 folder13 folder15 folder2 folder4 folder6 folder8
folder10 folder12 folder14 folder16 folder3 folder5 folder7 folder9
root@client1:/OG_Dataset# ls Dataset2
folder1 folder2 folder3 folder4 folder5 folder6 folder7 folder8
root@client1:/OG_Dataset# ls Dataset3
folder1 folder2 folder3 folder4 folder5 folder6 folder7 folder8
root@client1:/OG_Dataset# ls Dataset4
folder1 folder2 folder3 folder4
root@client1:/OG_Dataset# ls Dataset5
folder1 folder2 folder3 folder4
```

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```
root@client2:/OG_Dataset# ls
Dataset2 Dataset3 Dataset4 Dataset5
root@client2:/OG_Dataset# ls Dataset2
folder10 folder11 folder12 folder13 folder14 folder15 folder16 folder9
root@client2:/OG_Dataset# ls Dataset3
folder10 folder11 folder12 folder13 folder14 folder15 folder16 folder9
root@client2:/OG_Dataset# ls Dataset4
folder5 folder6 folder7 folder8
root@client2:/OG_Dataset# ls Dataset5
folder5 folder6 folder7 folder8
```

```
root@client3:/OG_Dataset# ls
Dataset4 Dataset5
root@client3:/OG_Dataset# ls Dataset4
folder10 folder11 folder12 folder9
root@client3:/OG_Dataset# ls Dataset5
folder10 folder11 folder12 folder9
```

```
root@client4:/OG_Dataset# ls
Dataset4 Dataset5
root@client4:/OG_Dataset# ls Dataset4
folder13 folder14 folder15 folder16
root@client4:/OG_Dataset# ls Dataset5
folder13 folder14 folder15 folder16
```

```
cc@pa4-kedar-extracredit:~$ sudo lxc launch ubuntu:22.04 server -c "limits.cpu=4" -c "limits.memory=2GiB"
Creating server
Starting server
cc@pa4-kedar-extracredit:~$ |
```

```
cc@pa4-kedar-extracredit:~$ for i in {1..4}; do
    sudo lxc launch ubuntu:22.04 client$i -c "limits.cpu=2" -c "limits.memory=2GiB"
done
Creating client1
Starting client1
Creating client2
Starting client2
Creating client3
Starting client3
Creating client4
Starting client4
cc@pa4-kedar-extracredit:~$
```

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```
cc@pa4-kedar-extracredit:~$ lxc list
To start your first container, try: lxc launch ubuntu:22.04
Or for a virtual machine: lxc launch ubuntu:22.04 --vm
```

NAME	STATE	IPV4	IPV6	TYPE	SNAPSHOTS
client1	RUNNING		fd42:baf7:6882:f302:216:3eff:fe53:e2e7 (eth0)	CONTAINER	0
client2	RUNNING		fd42:baf7:6882:f302:216:3eff:fe37:eea0 (eth0)	CONTAINER	0
client3	RUNNING		fd42:baf7:6882:f302:216:3eff:feb4:e76e (eth0)	CONTAINER	0
client4	RUNNING		fd42:baf7:6882:f302:216:3eff:fef4:60e1 (eth0)	CONTAINER	0
server	RUNNING		fd42:baf7:6882:f302:216:3eff:feaa:6ab9 (eth0)	CONTAINER	0

```
cc@pa4-kedar-extracredit:~$
```

End of the container part

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CSC435 Extra Credit Assignment - I

Chameleon resource used: compute_cascadelake_r

Virtual Machines

First I'll answer the questions and then I'll write commands/paste the command(s) images.

1. **Virtual machine technology used:** LXD

a. Command for **LXD** installation: `sudo snap install lxd`

2. **5 Virtual machines creation:** First I have created a storage pool to preallocate space for all clients and the server. After creating the pool I have used some commands for creation of the virtual machines.

a. `lxc storage create mypool zfs size=350GB`

b. `lxc init ubuntu:22.04 server --vm -c limits.cpu=4 -c limits.memory=2GB -s 60GB --storage mypool`

c. `lxc init ubuntu:22.04 client1 --vm -c limits.cpu=2 -c limits.memory=2GB -s 60GB --storage mypool`

d. `lxc init ubuntu:22.04 client2 --vm -c limits.cpu=2 -c limits.memory=2GB -s 60GB --storage mypool`

e. `lxc init ubuntu:22.04 client3 --vm -c limits.cpu=2 -c limits.memory=2GB -s 60GB --storage mypool`

f. `lxc init ubuntu:22.04 client4 --vm -c limits.cpu=2 -c limits.memory=2GB -s 60GB --storage mypool`

3. **All containers on Same network:** I have attached all the vm's to a common bridge network.

a. So I have created a network bridge.

b. Then I have assigned all the vm's to the bridge using the following command:

i. `lxc network attach bridge_testing client1`

4. **Copying the datasets to the clients:**

a. One example of copying is below.

i. **winscp@instance:** `scp -r scp -r /mnt/d/Depaul_Current/Distributed_Systems_1/Assignment_2/_Work /csc435-pa2-KedarChandulur/datasets/Zipped/Dataset4_Third_Quarter.zip cc@129.114.108.70:~/my_mounting_point/temp/`

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CSC435 Extra Credit Assignment - I

- ii. chameleon@instance: lxc file push
~/my_mounting_point/temp/Dataset4_Third_Quarter.zip
client3/Datasets_VM/
 - iii. client1@instance: unzip Dataset4_Third_Quarter.zip Dataset4
5. **OPTIONAL: Running the programs on the containers:** Not done.
 6. **Most challenging aspect of configuring containers:** Most challenging aspect was creating the first virtual machine and connecting it to a network.(The process of connecting to the network was pretty much the same as containers but for some reason when I was initially trying to do it the chameleon instance was kind of went to non-responsive state multiple times.)
 7. **Additional note:** *Almost all the commands were exactly same like for creating the containers and vm's, exlucding the vm creation command. The one and only difference which I have found is adding -vm for creating the vm and also using init instead of launch.*

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From here, I'm posting all the screen shots of the commands

```
root@client1:/Datasets_VM# ls Dataset1
folder1  folder11  folder13  folder15  folder2  folder4  folder6  folder8
folder10 folder12  folder14  folder16  folder3  folder5  folder7  folder9
root@client1:/Datasets_VM# ls Dataset2
folder1  folder2  folder3  folder4  folder5  folder6  folder7  folder8
root@client1:/Datasets_VM# ls Dataset3
folder1  folder2  folder3  folder4  folder5  folder6  folder7  folder8
root@client1:/Datasets_VM# ls Dataset4
folder1  folder2  folder3  folder4
root@client1:/Datasets_VM# ls Dataset5
folder1  folder2  folder3  folder4
root@client1:/Datasets_VM#
```

```
root@client2:/Datasets_VM# ls Dataset2
folder10 folder11 folder12 folder13 folder14 folder15 folder16 folder9
root@client2:/Datasets_VM# ls Dataset3
folder10 folder11 folder12 folder13 folder14 folder15 folder16 folder9
root@client2:/Datasets_VM# ls Dataset4
folder5  folder6  folder7  folder8
root@client2:/Datasets_VM# ls Dataset5
folder5  folder6  folder7  folder8
root@client2:/Datasets_VM#
```

```
root@client3:/Datasets_VM# ls Dataset4
folder10 folder11 folder12 folder9
root@client3:/Datasets_VM# ls Dataset5
folder10 folder11 folder12 folder9
root@client3:/Datasets_VM#
```

```
root@client4:/Datasets_VM# ls Dataset4
folder13 folder14 folder15 folder16
root@client4:/Datasets_VM# ls Dataset5
folder13 folder14 folder15 folder16
```

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